

STATISTICAL PHARMACOLOGY VIA KAKUTANI DICHOTOMY IX: NOTCH-SPARING γ -SECRETASE MODULATORS FOR ALZHEIMER'S DISEASE, SCHUR-COMPLEMENT CONFINEMENT OF THE CATALYTIC BLOCK, AND A CONE-GEODESIC MODEL OF PROCESSIVE TRIMMING

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SUMMARY. Papers I–VIII built the Kakutani indices and their estimators. This paper is the first application of the machinery to a single pharmacological target: the presenilin-1 catalytic core of γ -secretase, whose orthosteric inhibition failed in phase III because the enzyme also cleaves Notch. We ask for a ligand in quadrant II of the four-quadrant classifier of Paper IV: silent at the active site, loud in the second-order index. (1) We build a harmonic model of the catalytic core with $D = 300$ coordinates in three blocks, a catalytic block $\mathcal{C}$ of 30, a TM6a/PAL gate $\mathcal{G}$ of 70 and a TM1–TM9 channel $\mathcal{H}$ of 200. Its stiffness is a weighted graph Laplacian plus a bilayer tether, so Proposition 2.2 gives $\lambda_{\min} = t = 0.9000$ exactly, with condition number 27.62. (2) Theorem 2.3 is the mechanism of quadrant II. Because the catalytic block meets the trimming machinery $\mathcal{M} = \mathcal{G} \cup \mathcal{H}$ only through a weak coupling block $K_{\mathcal{CM}}$, any stiffness perturbation supported on $\mathcal{M}$ changes the catalytic precision matrix by an amount that is exactly bilinear in $K_{\mathcal{CM}}$; the induced mean displacement is first order and the induced index is therefore second order in the coupling, its covariance part fourth order. A scan over a 64-fold range of the coupling scale s reproduces the predicted exponents: 1.9906, 1.9709, 1.9755 and 3.9633 against 2, 2, 2 and 4, with successive local exponents converging to 2 and 4, while CKI_{TM} moves by 0.50%. (3) Theorem 3.2 solves the discrete infinity-Laplacian cone geodesic on the three-residue step lattice in closed form for equal weights, $d_L^{\text{cone}} = w \sum_{m \leq k} m^{-1/2}$, reproduced to 4.44×10^{-16} for $k \leq 6$, with $\max_j w_j \leq d_L^{\text{cone}} < \sum_j w_j$ for $k \geq 2$. Proposition 3.3 shows the anisotropy family $w(a) = \bar{w}(1 + a, 1, 1 - a)$ is even in a , so the isotropic point is critical; on 2001 grid points it is the strict minimiser, with minimum increment 6.71×10^{-8} . Cooperativity must be shared across all three helices to be worth anything. (4) Against this model, Semagacestat lands in quadrant IV ($K_{\text{catalytic}} = 6.72 \times 10^{-1}$, Notch retention 13.31%, $\text{CKI}_{\text{TM}} = 0.0344$, ratio unchanged at 1.004 $\times$), Flurbiprofen in quadrant III (safe, $\text{CKI}_{\text{TM}} = 0.2194$, 1.021 $\times$), and the designed lead KMS-AD-309 in quadrant II ($K_{\text{catalytic}} = 1.69 \times 10^{-3}$, retention 99.49%, $\text{CKI}_{\text{TM}} = 4.2536$ of which 95.11% is the eigenvector-rotation part, $A\beta_{42/40}$ down 2.61 $\times$). Two independent routes agree to a factor 1.34. (5) Proposition 4.1 shows the selectivity index saturates at $\text{CKI}_{\text{TM}}/\varepsilon_0$, which is why Flurbiprofen reaches 93% of its ceiling while failing the potency constraint outright. Proposition 4.2 separates what is definitional from what is empirical in the observation that a conventional activity assay returns 120 hits out of 240 ligands with 0% recall on quadrant II. The dynamics are a harmonic surrogate on a schematic contact graph, the compounds enter only through their ligand class, and KMS-AD-309 is a designed point in parameter space. Section 6 states these limits and five open problems.

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<https://github.com/Ruqing1963/kakutani-statistical-pharmacology-IX>. Papers I–VIII of the series are [5, 6, 7, 8, 9, 10, 11, 12], the integrating article is [13], and the underlying monograph is [4].

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1. INTRODUCTION: THE QUADRANT THAT THE ASSAY CANNOT SEE

1.1. A target that punishes potency. γ -Secretase is an intramembrane aspartyl protease whose catalytic subunit is presenilin-1. It performs the final cut that releases the amyloid- β peptides from the C-terminal fragment of the amyloid precursor protein, and it does so processively: after an initial ϵ -cut it trims the substrate in steps of three residues, $A\beta_{49} \rightarrow A\beta_{46} \rightarrow A\beta_{43} \rightarrow A\beta_{40}$ along one product line and $A\beta_{48} \rightarrow A\beta_{45} \rightarrow A\beta_{42} \rightarrow A\beta_{38}$ along the other. What matters clinically is not how much $A\beta$ the enzyme makes but which species it releases, because the aggregation propensity of the products is steeply graded: $A\beta_{42}$ nucleates, $A\beta_{40}$ and $A\beta_{38}$ largely do not. The pathogenic observable is a ratio, not a rate.

The same enzyme also performs the S3 cut that liberates the Notch intracellular domain. An orthosteric inhibitor occupies the catalytic aspartate dyad, D257 on TM6 and D385 on TM7, and therefore suppresses both substrates at once. That is the arithmetic behind the failure of Semagacestat: a compound that inhibited γ -secretase exactly as designed, and whose phase III trials were stopped for worsened cognition and for toxicities consistent with loss of Notch signalling. Reducing total $A\beta$ by shutting the enzyme down is not the same

intervention as shifting the product distribution, and the clinical record has been unusually clear about the difference.

What is wanted instead is a *modulator*: a ligand that leaves the dyad and the S3 register geometrically undisturbed and instead changes the register of processive trimming, so that the $A\beta_{42}$ line is pushed onward to $A\beta_{38}$. The first generation of such compounds, derived from non-steroidal anti-inflammatory drugs, was safe and too weak; R-flurbiprofen reached phase III and failed on efficacy. Safety and potency here are not a trade-off along one axis. They are two coordinates.

1.2. The four-quadrant reading. Paper IV introduced a classifier for ligands in the plane spanned by a local, pocket-visible perturbation and the second-order correlated Kakutani index CKI , and observed that the pharmacologically interesting class sits where the first coordinate is small and the second is large: the ligand is invisible to a screen that watches the binding site, and loud in the covariance of the distal ensemble. Papers V–VIII supplied the obstruction group, the cone-geodesic collective moves, the estimator pipeline `kakutani_pharma`, and the finite-sample calibration. All of that was developed on synthetic or generic systems.

This paper puts the machinery on one target and asks it to reproduce a known clinical separation. The two coordinates become concrete:

- $K_{\text{catalytic}}(L)$, the correlated Kakutani index between the apo and liganded ensembles restricted to the 30 coordinates of the catalytic dyad and the Notch S3 register. This is the Notch-safety coordinate, and the design brief asks for $K_{\text{catalytic}} < 0.025$.
- $\text{CKI}_{\text{TM}}(L)$, the same index on the 270 coordinates of the trimming machinery, the TM6a/PAL gate together with the TM1–TM9 channel. This is the potency coordinate, and the brief asks for $\text{CKI}_{\text{TM}} > 3.2$.

Quadrant II is the conjunction. Quadrant IV, high $K_{\text{catalytic}}$ and low CKI_{TM} , is the orthosteric inhibitor; quadrant III, both low, is the silent analogue; quadrant I, both high, is a potent ligand that has not escaped the Notch liability.

1.3. Three results. The paper contributes three things, in increasing order of what they cost to establish.

First, a *mechanism* for why quadrant II is reachable at all. Section 2 builds a three-zone harmonic model of the catalytic core and Theorem 2.3 shows that the catalytic block is screened from the machinery at second order in the inter-block coupling. The screening is not an artefact of the perturbation being small: the perturbation is large, $\text{CKI}_{\text{TM}} = 4.25$ against a threshold of 3.2, while what reaches the catalytic block is $K_{\text{catalytic}} = 1.7 \times 10^{-3}$. The Schur complement is the reason. A scan over the coupling scale confirms the predicted exponents to two decimal places.

Second, a *geometry* for the three-residue step. Section 3 treats the synchronous displacement of TM3, TM6a and the PAL motif as a move from $(0, 0, 0)$ to $(1, 1, 1)$ on a three-bit lattice and computes its cost with the discrete infinity-Laplacian cone distance of Paper VI. Theorem 3.2 gives the closed form for equal weights, $d_L^{\text{cone}} = w(1 + 2^{-1/2} + 3^{-1/2}) = 2.284457 w$, strictly below the sequential cost $3w$; the discount is what a modulator buys by making the three displacements cooperative. Proposition 3.3 and Numerical observation 3.4 show that the discount is maximised exactly at the isotropic point, so a ligand that couples one helix strongly and the others weakly pays nearly the sequential price.

Third, a *calibration* against the clinical record. Section 4 places Semagacestat, Flurbiprofen and a designed lead KMS-AD-309 in the plane, screens 240 ligands, and examines what a conventional catalytic activity read-out does with them. We are careful in Proposition 4.2 to separate the part of that observation which is true by definition from the part which is an empirical property of the model.

1.4. What this paper does not do. No structure is docked, no free energy is computed, and no molecule is synthesised. The dynamics are Gaussian and harmonic on a contact graph with the correct block topology, not molecular dynamics; the three compounds enter through

three occupancy strengths rather than through their chemistry; and KMS-AD-309 is a point in that parameter space. The model reproduces a separation that is already known; it does not predict it from structure. Section 6 states this in detail, and everything in the paper is generated by two scripts whose acceptance tests are reproduced in Section 5.

2. A THREE-ZONE GAUSSIAN ELASTIC MODEL AND SCHUR-COMPLEMENT CONFINEMENT

2.1. The model.

Definition 2.1 (Three-zone harmonic model). Let V be a set of $D = 300$ harmonic degrees of freedom partitioned into fourteen segments, which are grouped into three blocks:

$$\begin{aligned}\mathcal{C} &= \text{D257 loop (8)} \cup \text{D385 loop (8)} \cup \text{S3 register (14)}, & |\mathcal{C}| &= 30, \\ \mathcal{G} &= \text{TM6a (40)} \cup \text{PAL (30)}, & |\mathcal{G}| &= 70, \\ \mathcal{H} &= \text{TM1 (24)}, \dots, \text{TM9 (16)}, & |\mathcal{H}| &= 200,\end{aligned}$$

and write $\mathcal{M} = \mathcal{G} \cup \mathcal{H}$ for the trimming machinery, $|\mathcal{M}| = 270$. Let $G(L)$ be a weighted graph on V whose edge weights depend on a ligand L , and let

$$K(L) = \Lambda(G(L)) + tI_D, \quad t = 0.9, \quad (2.1)$$

where Λ is the weighted graph Laplacian. The conformational ensemble of L is the Gaussian $N(\mu(L), \Sigma(L))$ with

$$\Sigma(L) = K(L)^{-1}, \quad \mu(L) = \Sigma(L) f(L), \quad (2.2)$$

in units of $k_B T$, where $f(L)$ is the ligand binding force.

The edges of G are of three kinds. Inside each segment a backbone path carries weight 6.0, 4.0 or 5.0 according to the block. Between segments there are packing contacts: nine consecutive helix-helix pairs plus three closures at weight 1.2 inside the channel, TM6a–PAL at 1.5, four gate-channel contacts at 1.0, and three internal catalytic contacts at 0.8. Finally, and decisively, the catalytic block meets the rest of the structure only through D257–TM6 and D385–TM7 at weight 0.35 and S3 register–PAL at weight 0.25. The placement of the two aspartates on TM6 and TM7 follows presenilin-1; the weakness of the three couplings is the design hypothesis whose consequences Theorem 2.3 draws out.

A ligand acts through three occupancy strengths $s_{\text{cat}}, s_{\text{gate}}, s_{\text{chan}} \geq 0$. The first adds springs across the dyad and the S3 register; the second couples TM6a to PAL and softens the TM6a backbone; the third creates cross-helix couplings TM3–TM6, TM3–TM7, TM5–TM9, TM2–TM8 and, critically, TM3–TM6a, while softening the TM3 and TM6 backbones. Each also contributes a binding force supported on its own block. Softening factors are floored at 0.15, so every edge weight stays positive.

Proposition 2.2 (Positive definiteness and the exact spectral floor). *For every ligand, $K(L) \succ 0$ and $\lambda_{\min}(K(L)) = t$ exactly, attained on the uniform direction $D^{-1/2}\mathbf{1}$. In particular $\dim \ker K = 0$ and $\Sigma(L)$ is well defined.*

Proof. All edge weights are positive, so $\Lambda(G) \succeq 0$. The graph G is connected, hence $\ker \Lambda(G) = \mathbb{R}\mathbf{1}$ and the second eigenvalue is positive. Adding tI shifts the whole spectrum by t , so $\lambda_{\min}(K) = 0 + t = t$ with eigenvector $\mathbf{1}$, and every other eigenvalue exceeds t . $\square$

The numerical spectrum of the apo model is $[0.9000, 24.8622]$, condition number 27.62, in agreement with Proposition 2.2; see Table 1. The floor is attained, not approached, which is a useful invariant: any coding error that disturbed the Laplacian structure would move it.

2.2. Indices. Following Paper II, for two Gaussians the correlated Kakutani index is the Bhattacharyya distance

$$\text{CKI} = \underbrace{\frac{1}{8} \Delta\mu^\top \bar{\Sigma}^{-1} \Delta\mu}_{D_{\text{mean}}} + \underbrace{\frac{1}{2} \left[\log \det \bar{\Sigma} - \frac{1}{2} (\log \det \Sigma_A + \log \det \Sigma_B) \right]}_{D_{\text{cov}}}, \quad \bar{\Sigma} = \frac{1}{2} (\Sigma_A + \Sigma_B), \quad (2.3)$$

and $D_{\text{cov}} = D_{\text{comm}} + D_{\text{rot}}$ splits into a commuting spectral part and a non-negative eigenvector-rotation part. We write $K_{\text{catalytic}}(L)$ for (2.3) evaluated on the marginals over $\mathcal{C}$, and $\text{CKI}_{\text{TM}}(L)$ for the same on $\mathcal{M}$; $\text{CKI}_{\text{cov}}(L)$ denotes the covariance part of $\text{CKI}_{\text{TM}}(L)$. A Gaussian marginal is the corresponding sub-block of Σ , so these are computed without any further approximation.

The Notch read-out is defined as follows. The S3 cut requires the correct register at three consecutive subsites, and we take each subsite to retain its register with probability equal to the Bhattacharyya affinity $e^{-K_{\text{catalytic}}}$ of the catalytic ensembles. Hence

$$\text{Notch retention} = e^{-3K_{\text{catalytic}}}, \quad (2.4)$$

so that the design threshold $K_{\text{catalytic}} < 0.025$ is exactly the statement that retention exceeds $e^{-0.075} = 0.9277$.

2.3. Schur-complement confinement. Write the apo stiffness in block form and let P denote the precision matrix of the catalytic marginal,

$$K = \begin{pmatrix} K_{\mathcal{C}\mathcal{C}} & K_{\mathcal{C}\mathcal{M}} \\ K_{\mathcal{M}\mathcal{C}} & K_{\mathcal{M}\mathcal{M}} \end{pmatrix}, \quad P = \Sigma_{\mathcal{C}\mathcal{C}}^{-1} = K_{\mathcal{C}\mathcal{C}} - K_{\mathcal{C}\mathcal{M}}K_{\mathcal{M}\mathcal{M}}^{-1}K_{\mathcal{M}\mathcal{C}}. \quad (2.5)$$

An allosteric ligand, by definition, perturbs only the machinery: ΔK is supported on $\mathcal{M} \times \mathcal{M}$ and f is supported on $\mathcal{M}$.

Theorem 2.3 (Second-order screening of the catalytic block). *Let ΔK be symmetric, supported on $\mathcal{M} \times \mathcal{M}$, with $K + \Delta K \succ 0$, and let f be supported on $\mathcal{M}$. Then:*

(i) *The catalytic precision changes by*

$$\Delta P = -K_{\mathcal{C}\mathcal{M}} \left[(K_{\mathcal{M}\mathcal{M}} + \Delta K_{\mathcal{M}\mathcal{M}})^{-1} - K_{\mathcal{M}\mathcal{M}}^{-1} \right] K_{\mathcal{M}\mathcal{C}}, \quad (2.6)$$

which is exactly bilinear in the coupling block $K_{\mathcal{C}\mathcal{M}}$. Consequently, if $K_{\mathcal{C}\mathcal{M}}$ is replaced by $sK_{\mathcal{C}\mathcal{M}}$ while $K_{\mathcal{C}\mathcal{C}}$ and $K_{\mathcal{M}\mathcal{M}}$ are held fixed, then $\Delta P = s^2 \Delta P_1$ with ΔP_1 independent of s , and $\|\Delta \Sigma_{\mathcal{C}\mathcal{C}}\| = O(s^2)$.

(ii) *The catalytic mean displacement is $\mu_{\mathcal{C}} = -\Sigma_{\mathcal{C}\mathcal{C}}K_{\mathcal{C}\mathcal{M}}K_{\mathcal{M}\mathcal{M}}^{-1}f_{\mathcal{M}} = O(s)$.*

(iii) *Therefore $D_{\text{mean}}(K_{\text{catalytic}}) = O(s^2)$ and $D_{\text{cov}}(K_{\text{catalytic}}) = O(s^4)$, so $K_{\text{catalytic}} = O(s^2)$, dominated at small s by the mean term.*

(iv) *The index on the machinery, CKI_{TM} , is unaffected to leading order: it depends on $K_{\mathcal{C}\mathcal{M}}$ only through $K_{\mathcal{M}\mathcal{M}} - K_{\mathcal{M}\mathcal{C}}K_{\mathcal{C}\mathcal{C}}^{-1}K_{\mathcal{C}\mathcal{M}}$, again an $O(s^2)$ correction to an $O(1)$ quantity.*

Proof. (i) An $\mathcal{M}$ -supported perturbation leaves $K_{\mathcal{C}\mathcal{C}}$ and $K_{\mathcal{C}\mathcal{M}}$ untouched, so subtracting the two instances of (2.5) cancels $K_{\mathcal{C}\mathcal{C}}$ and leaves (2.6). The bracket depends on s not at all, and $K_{\mathcal{C}\mathcal{M}}$ appears once on each side, giving the exact factor s^2 . Since $\Sigma_{\mathcal{C}\mathcal{C}} = P^{-1}$ and $P \succ 0$ uniformly, $\Delta \Sigma_{\mathcal{C}\mathcal{C}} = -P^{-1}\Delta P(P + \Delta P)^{-1} = O(\|\Delta P\|) = O(s^2)$.

(ii) The block-inverse formula gives $(K^{-1})_{\mathcal{C}\mathcal{M}} = -\Sigma_{\mathcal{C}\mathcal{C}}K_{\mathcal{C}\mathcal{M}}K_{\mathcal{M}\mathcal{M}}^{-1}$, and $\mu_{\mathcal{C}} = (K^{-1}f)_{\mathcal{C}} = (K^{-1})_{\mathcal{C}\mathcal{M}}f_{\mathcal{M}}$ because $f_{\mathcal{C}} = 0$. One factor of $K_{\mathcal{C}\mathcal{M}}$ appears, so $\mu_{\mathcal{C}} = O(s)$.

(iii) In the apo state $f = 0$, hence $\mu_{\mathcal{C}}^{\text{apo}} = 0$ and $\Delta \mu_{\mathcal{C}} = \mu_{\mathcal{C}} = O(s)$. The mean term of (2.3) is a quadratic form in $\Delta \mu_{\mathcal{C}}$ with a bounded kernel, whence $O(s^2)$. For the covariance term, set $E = \Delta \Sigma_{\mathcal{C}\mathcal{C}} = O(s^2)$ and expand (2.3) about $E = 0$: the first-order terms cancel identically, since $\log \det$ is concave and the expression is a symmetrised Bregman divergence, leaving $D_{\text{cov}} = \frac{1}{8} \text{tr}[(\Sigma_{\mathcal{C}\mathcal{C}}^{-1}E)^2] + O(\|E\|^3) = O(s^4)$. Adding the two gives $K_{\text{catalytic}} = O(s^2)$.

(iv) Symmetrically to (2.5), the marginal precision on $\mathcal{M}$ is $K_{\mathcal{M}\mathcal{M}} - K_{\mathcal{M}\mathcal{C}}K_{\mathcal{C}\mathcal{C}}^{-1}K_{\mathcal{C}\mathcal{M}}$, in which $K_{\mathcal{C}\mathcal{M}}$ again occurs twice. $\square$

Remark 2.4. The content of Theorem 2.3 is not that the perturbation is small. It is that the catalytic block sees the machinery only through a bilinear form in the coupling, so a designer has a quadratic lever: halving the catalytic coupling divides the leakage by four while leaving the potency coordinate where it was. This is the quantitative version of the informal statement that an allosteric site must be *dynamically* as well as spatially distant from the active site.

Corollary 2.5 (Quadratic growth of selectivity, and its ceiling). *With $\varepsilon_0 > 0$ as in (4.1) below, the selectivity index of a purely allosteric ligand satisfies $S_{\text{Notch}}(s) = \text{CKI}_{\text{TM}}/(K_{\text{catalytic}}(s) + \varepsilon_0) \sim Cs^{-2}$ while $K_{\text{catalytic}}(s) \gg \varepsilon_0$, and saturates at $\text{CKI}_{\text{TM}}/\varepsilon_0$ once $K_{\text{catalytic}}(s) \ll \varepsilon_0$.*

2.4. Numerical verification of the exponents. Table 2 scans the two catalytic couplings by a common factor s over a 64-fold range, with the perturbation held fixed at the purely allosteric ligand $s_{\text{cat}} = 0$, $s_{\text{gate}} = s_{\text{chan}} = 1$. Fitting on the four smallest scales gives exponents

$$\|\Delta P\|_F : 1.9906, \quad \text{leakage} : 1.9709, \quad D_{\text{mean}} : 1.9755, \quad D_{\text{cov}} : 3.9633,$$

against the predicted 2, 2, 2 and 4. The successive local exponents converge monotonically to those values as $s \rightarrow 0$, which is what an asymptotic statement should do:

$$\begin{aligned} \|\Delta P\|_F : & \quad 1.996, \quad 1.992, \quad 1.984, \quad 1.968, \quad 1.937, \quad 1.880, \\ D_{\text{cov}} : & \quad 3.990, \quad 3.966, \quad 3.933, \quad 3.870, \quad 3.753, \quad 3.552. \end{aligned}$$

Over the same scan CKI_{TM} varies by a factor 1.0050, confirming part (iv).

Two consequences deserve emphasis. At the design value $s = 1$ the purely allosteric ligand has $K_{\text{catalytic}} = 7.45 \times 10^{-7}$, which is 1342 times below ε_0 : the selectivity index of a clean allosteric ligand is already saturated. The lead KMS-AD-309 has $K_{\text{catalytic}} = 1.69 \times 10^{-3}$, a factor 2265 larger, and that excess is entirely due to its small residual active-site contact $s_{\text{cat}} = 0.02$, not to leakage. In other words, the Schur screening is not the binding constraint on this lead; the chemistry of its active-site contact is.

2.5. Spectral anatomy of the perturbation. The decomposition $D_{\text{cov}} = D_{\text{comm}} + D_{\text{rot}}$ of Paper II separates two things a ligand can do to a covariance: rescale the variances of the existing modes, which is the commuting part, or turn the modes themselves, which is the rotation part. The second is non-negative and vanishes exactly when the two covariances commute. Paper II argued that cryptic allostery is a rotation phenomenon, and the present model lets that be checked compound by compound. On the machinery block,

	D_{comm}	D_{rot}	$D_{\text{rot}}/D_{\text{cov}}$
Semagacestat	0.00572	0.02863	83.35 %
Flurbiprofen	0.02745	0.19189	87.49 %
KMS-AD-309	0.20776	4.04329	95.11 %

so all three act mainly by rotation, but the lead does so both absolutely and in share. Its commuting part, 0.208, is by itself an order of magnitude below the potency threshold: a ligand that only stiffened the channel uniformly, however hard it stiffened, would not reach quadrant II in this model. The eigenvectors have to move.

This is worth separating from the Schur result. Theorem 2.3 says a perturbation of the machinery need not reach the catalytic block. Section 2.5 says that the perturbation which does reach the *trimming* well enough to matter is of a specific spectral kind. The two together are the design target: rotate the machinery’s modes, and do it through couplings the catalytic block barely feels.

2.6. The mean term is negligible everywhere. One further diagnostic is worth recording because it is easy to misread. On the machinery block the mean term of (2.3) is 6.9×10^{-6} , 5.5×10^{-5} and 2.6×10^{-3} for the three compounds, that is, between 0.02 % and 0.06 % of the total index. The signal is essentially pure covariance. This matches the covariance-dominance theorem of Paper II and means that CKI_{TM} here is not measuring how far the ligand displaces the channel, but how much it reorganises the channel’s fluctuations: a displacement-based descriptor, such as a root-mean-square deviation, would register almost nothing. That is the formal content of the word “cryptic”.

3. THE CONE GEODESIC OF THE THREE-RESIDUE STEP AND ISOTROPIC OPTIMALITY

3.1. The step lattice and its distance. Processive trimming advances the substrate three residues at a time. In the model this requires the simultaneous displacement of three collective coordinates, x_{TM3} , x_{TM6a} and x_{PAL} , from $(0, 0, 0)$ to $(1, 1, 1)$. Paper VI showed that the cost of such a collective move is governed not by the sum of the individual costs but by the discrete infinity-Laplacian distance on the corresponding lattice.

Definition 3.1 (Cone distance on the cube). Fix $w \in (0, \infty)^k$. For $x \in \{0, 1\}^k$ let $F(x) = \{j : x_j = 0\}$. Define $h : \{0, 1\}^k \rightarrow [0, \infty)$ by $h(\mathbf{1}) = 0$ and, for $x \neq \mathbf{1}$, by taking $h(x)$ to be the root in $[\max_{j \in F(x)} h(x + e_j), \infty)$ of

$$\sum_{j \in F(x)} \left(\frac{t - h(x + e_j)}{w_j} \right)^2 = 1. \quad (3.1)$$

Set $d_L^{\text{cone}}(w) = h(\mathbf{0})$.

Equation (3.1) is the discrete eikonal, or Aronsson, condition: the forward difference vector, measured in the w -scaled Euclidean norm, has unit length. The left-hand side is continuous and strictly increasing in t on the stated interval, so the root is unique *provided* the value at the left endpoint does not exceed 1, that is provided

$$g(x) = 1 - \sum_{j \in F(x)} \left(\frac{\max_i h(x + e_i) - h(x + e_j)}{w_j} \right)^2 \geq 0. \quad (3.2)$$

This is a hypothesis on w , not a theorem. We verify it: the minimum of g over all vertices is 0.979027 over the 243 ligands studied here, and 7.2×10^{-5} over the anisotropy family of Section 3.3, where it degenerates only as one weight tends to zero. The degenerate case is listed among the open problems.

Theorem 3.2 (Closed form and elementary bounds). *Let $k \geq 1$ and $w \in (0, \infty)^k$.*

- (i) (Homogeneity and symmetry.) $d_L^{\text{cone}}(\lambda w) = \lambda d_L^{\text{cone}}(w)$ for $\lambda > 0$, and d_L^{cone} is invariant under permutations of the coordinates of w .
- (ii) (Equal weights.) If $w_j = w$ for all j , then

$$d_L^{\text{cone}}(w, \dots, w) = w \sum_{m=1}^k \frac{1}{\sqrt{m}}. \quad (3.3)$$

For $k = 3$ this is $w(1 + 2^{-1/2} + 3^{-1/2}) = 2.284457 w$.

- (iii) (Bounds.) $\max_j w_j \leq d_L^{\text{cone}}(w)$, and $d_L^{\text{cone}}(w) < \sum_j w_j$ whenever $k \geq 2$.

Proof. (i) Replacing w by λw in (3.1) and substituting $t = \lambda t'$, $h = \lambda h'$ reproduces the same system, and the recursion is manifestly invariant under the lattice automorphism induced by a permutation of coordinates.

(ii) Downward induction on $m = |F(x)|$. For $m = 1$, say $F(x) = \{j\}$, (3.1) reads $((t - 0)/w)^2 = 1$, so $h(x) = w = D_1$, where $D_m := w \sum_{r \leq m} r^{-1/2}$. Assume $h(y) = D_{m-1}$ for every y with $|F(y)| = m - 1$. For x with $|F(x)| = m$, every neighbour $x + e_j$, $j \in F(x)$, has $|F| = m - 1$, so all m terms of (3.1) share the same h -value D_{m-1} , and the equation becomes $m((t - D_{m-1})/w)^2 = 1$, whose root above D_{m-1} is $t = D_{m-1} + w/\sqrt{m} = D_m$. Taking $m = k$ and $x = \mathbf{0}$ gives (3.3). Note that with all h -values equal, $g(x) = 1 > 0$, so (3.2) holds automatically.

(iii) Lower bound, by downward induction on $|F(x)|$: the claim is $h(x) \geq \max_{i \in F(x)} w_i$. For $|F(x)| = 1$ it is an equality. For $|F(x)| = m \geq 2$ fix $i \in F(x)$ and choose any $j \in F(x) \setminus \{i\}$; then $i \in F(x + e_j)$, so by induction $h(x + e_j) \geq w_i$, and $h(x) \geq \max_{j \in F(x)} h(x + e_j) \geq w_i$. As i was arbitrary, $h(x) \geq \max_{i \in F(x)} w_i$.

Upper bound, again by downward induction, with the claim $h(x) < \sum_{i \in F(x)} w_i$ for $|F(x)| \geq 2$ and $h(x) = w_j$ for $F(x) = \{j\}$. For $|F(x)| = 2$, say $F(x) = \{1, 2\}$, the neighbours give

$h(x + e_1) = w_2$ and $h(x + e_2) = w_1$, and at $t = w_1 + w_2$ the left-hand side of (3.1) equals $1 + 1 = 2 > 1$; since it is increasing, the root satisfies $h(x) < w_1 + w_2$. For $|F(x)| = m \geq 3$, every term of (3.1) is at most 1, so $h(x) \leq h(x + e_j) + w_j$ for each $j \in F(x)$; choosing any such j and applying the inductive bound $h(x + e_j) < \sum_{i \in F(x) \setminus \{j\}} w_i$ gives $h(x) < \sum_{i \in F(x)} w_i$. $\square$

The recursion of Definition 3.1 is implemented exactly, and reproduces (3.3) for $k = 1, \dots, 6$ with a maximum absolute error of 4.44×10^{-16} , one unit in the last place. The bounds of (iii) hold in all 243 evaluations reported here. For the step weight $w_0 = 1.05 k_B T$ used below,

$$d_L^{\text{cone}} = 2.398680 k_B T \quad \text{against} \quad 3w_0 = 3.150000 k_B T, \quad \frac{d_L^{\text{cone}}}{3w_0} = 0.761486.$$

3.2. Step weights from the model. The weights are not free parameters. For each of the three step segments let e_j be the unit vector of a uniform displacement of that segment. The variance of the collective coordinate with everything else relaxed is $e_j^\top \Sigma e_j$, whose reciprocal is the effective free-energy curvature, and we set

$$w_j(L) = w_0 \sqrt{\frac{e_j^\top \Sigma(\text{apo}) e_j}{e_j^\top \Sigma(L) e_j}}, \quad w_0 = 1.05 k_B T, \quad (3.4)$$

so that $w_j \equiv w_0$ for the apo model by construction. A ligand that stiffens the step coordinates raises its w_j ; by homogeneity this raises d_L^{cone} and the sequential cost in the same proportion, so the *relative* discount depends only on the direction of w , not on its magnitude.

3.3. Isotropic optimality. The question is how the discount $\sum_j w_j - d_L^{\text{cone}}(w)$ depends on the direction of w at fixed total. Consider the one-parameter family

$$w(a) = \bar{w} (1 + a, 1, 1 - a), \quad a \in [0, 1), \quad (3.5)$$

which has $\sum_j w_j(a) = 3\bar{w}$ for every a .

Proposition 3.3 (Evenness and criticality). *$a \mapsto d_L^{\text{cone}}(w(a))$ is an even function of a on $(-1, 1)$. Consequently, if it is differentiable at $a = 0$, then $a = 0$ is a critical point.*

Proof. $w(-a) = \bar{w}(1 - a, 1, 1 + a)$ is the reversal of $w(a)$, hence a permutation of it, and d_L^{cone} is permutation invariant by Theorem 3.2(i). $\square$

Numerical observation 3.4 (Isotropy minimises the cone distance along the family). On a uniform grid of 2001 points $a \in [0, 0.999]$ with $\bar{w} = 1.05 k_B T$, the map $a \mapsto d_L^{\text{cone}}(w(a))$ is strictly increasing: the minimum successive increment is $6.712 \times 10^{-8} k_B T$, and the value rises from 2.398680 to 2.729042 $k_B T$. Hence within this family the isotropic point is the strict minimiser of d_L^{cone} and the strict maximiser of the discount, which falls from 0.751320 to 0.420958 $k_B T$.

We record this as a numerical observation rather than a theorem. The natural statement behind it is that d_L^{cone} is Schur-convex on the positive orthant; by symmetry, convexity would suffice, since for a symmetric convex function Jensen applied to the average over the $k!$ permutations gives $d_L^{\text{cone}}(\bar{w}\mathbf{1}) \leq d_L^{\text{cone}}(w)$ whenever $\sum_j w_j = k\bar{w}$. We have not proved convexity, and list it as Open problem (1).

The pharmacological reading is direct and, we think, not obvious in advance: cooperativity has to be spread. Table 3 shows the relative cost $d_L^{\text{cone}} / \sum_j w_j$ climbing from 0.761486 at $a = 0$ to 0.822364 at $a = 0.8$. A modulator that stiffens TM3 while leaving TM6a and the PAL motif loose does not convert the three-residue step into one cooperative move; it pays almost the sequential price, and the discount it buys is 44 % smaller than what an isotropic ligand of the same total stiffness would obtain.

3.4. From the discount to two observables. Only the fraction of steps that actually proceed synchronously travels the cone geodesic. We take that fraction to be

$$\sigma(L) = 1 - \exp(-\text{CKI}_{\text{TM}}(L)/\text{CKI}^*), \quad \text{CKI}^* = 3.20, \quad (3.6)$$

with CKI^* set equal to the potency threshold, so that a ligand which barely meets the design brief synchronises about 63 % of its steps. Two observables follow. The first is the transition-state discount for the $\text{A}\beta_{42} \rightarrow \text{A}\beta_{38}$ step,

$$\Delta\Delta G^\ddagger(L) = \sigma(L) k_B T \left[\sum_j w_j(L) - d_L^{\text{cone}}(w(L)) \right], \quad (3.7)$$

with $k_B T = 0.6162$ kcal/mol at 310 K. The second is the pathogenic ratio law of the design brief,

$$R_{42/40}(L) = R_0 \exp\left(-\eta[\text{CKI}_{\text{cov}}(L) - d_L^{\text{cone,eff}}(L)]\right), \quad d_L^{\text{cone,eff}}(L) = \sigma(L) d_L^{\text{cone}}(w(L)), \quad (3.8)$$

with $R_0 = 0.182$ and $\eta = 0.45$. The effective cone term is the residual cost of the synchronous step that the ligand must still pay, and it enters with a minus sign: the net driving force is the covariance rearrangement less that cost. The normalisation is such that the apo state, $\text{CKI}_{\text{cov}} = 0$ and $\sigma = 0$, returns R_0 exactly.

We stress what is and is not derived here. Equation (3.7) follows from Theorem 3.2 and the definition of σ . Equation (3.8) is the phenomenological law of the design brief; R_0 and η are calibrated, not derived, and this is Limit (d) of Section 6. What makes the pair informative is that they are not the same computation, so their agreement is a check. For the lead the ratio law predicts a $2.608\times$ fall in $R_{42/40}$, and the cone barrier (3.7) predicts a rate acceleration $\exp(\Delta\Delta G^\ddagger/k_B T) = 1.942\times$ for the trimming step. The quotient is 1.343.

The magnitude deserves a comment, because 0.409 kcal/mol looks small next to the barrier of a proteolytic step. It is small, and it should be. A product ratio is a difference of two barriers, not a barrier, and at 310 K a two-fold change in a branching ratio costs $k_B T \log 2 = 0.427$ kcal/mol. The quantity that (3.7) computes is therefore of exactly the right order for the effect being modelled, and a discount ten times larger would be a reason to distrust the model rather than to celebrate the lead.

3.5. How much of the discount do the compounds actually collect? Theorem 3.2 and Numerical observation 3.4 bound the relative discount from above by $1 - 0.761486 = 0.238514$, attained at isotropy. The three compounds sit at relative costs

$$d_L^{\text{cone}} / \sum_j w_j = 0.7614959, \quad 0.7615036, \quad 0.7616075$$

for Semagacestat, Flurbiprofen and KMS-AD-309 respectively, each within 1.5×10^{-4} of the isotropic optimum 0.7614857. The anisotropy that a ligand induces in the step weights through (3.4) is therefore, in this model, negligible: all three collect essentially the full geometric discount available, and the large differences in $\Delta\Delta G^\ddagger$ across Table 5 come from the synchronous fraction σ and from the overall scale of w , not from the shape of w .

That is a substantive finding and not a foregone one. It says that in this model the cone geometry contributes a nearly constant factor, and the whole of the pharmacological variation is carried by σ , that is, by CKI_{TM} . Anisotropy would begin to matter only for a ligand that stiffened one step helix several-fold relative to the others, which none of the three does; Table 3 shows that such a ligand would forfeit up to 44 % of the discount.

3.6. Relation to the single-step scan barrier of Paper VI. Paper VI introduced the cone geodesic as the cost of a collective move and contrasted it with the single-step scan barrier, the cost of the best sequential schedule. The present application is the smallest non-trivial instance, $k = 3$, and it makes the contrast operational: $\sum_j w_j$ is what the substrate pays if TM3, TM6a and the PAL motif move one after another, and d_L^{cone} is what it pays if a modulator has coupled them so that they move together. The ligand does not lower either cost directly; it changes which of the two applies, and the fraction of steps on which it does so

is σ . In this reading a γ -secretase modulator is a device for converting a sequential schedule into a synchronous one, and (3.7) is the value of that conversion.

4. THREE COMPOUNDS, A VIRTUAL SCREEN, AND WHAT AN ACTIVITY ASSAY SELECTS FOR

4.1. The selectivity index and its ceiling. The design brief asks for a single figure of merit,

$$S_{\text{Notch}}(L) = \frac{\text{CKI}_{\text{TM}}(L)}{K_{\text{catalytic}}(L) + \varepsilon_0}, \quad \varepsilon_0 = 10^{-3}. \quad (4.1)$$

Proposition 4.1 (Saturation). *For every ligand, $S_{\text{Notch}}(L) \leq \text{CKI}_{\text{TM}}(L)/\varepsilon_0$, with equality in the limit $K_{\text{catalytic}} \rightarrow 0$. Consequently:*

- (i) *on the set $\{K_{\text{catalytic}} \ll \varepsilon_0\}$ the index degenerates to a rescaling of the potency coordinate and carries no safety information at all;*
- (ii) *$S_{\text{Notch}}(L)$ large does not imply $\text{CKI}_{\text{TM}}(L) > \text{CKI}^*$; the two are related only through the factor $(1 + K_{\text{catalytic}}/\varepsilon_0)^{-1} \in (0, 1]$.*

Proof. Immediate from $K_{\text{catalytic}} \geq 0$. □

This is not a formal quibble. Flurbiprofen scores $S_{\text{Notch}} = 204.4$, which is 93.2% of its ceiling $\text{CKI}_{\text{TM}}/\varepsilon_0 = 219.4$, while its potency coordinate $\text{CKI}_{\text{TM}} = 0.2194$ is fifteen times below the threshold. Across all 243 ligands studied, the largest observed ratio $S_{\text{Notch}}\varepsilon_0/\text{CKI}_{\text{TM}}$ is 0.995727: the index is essentially always at its ceiling. A ranking index is what (4.1) provides; the two absolute constraints are what select. We recommend reporting S_{Notch} only alongside both.

4.2. Three compounds. Table 4 and Table 5 give the full read-out for the three reference ligands, whose occupancy strengths encode their pharmacological class and nothing else.

Semagacestat is modelled as a transition-state analogue: $s_{\text{cat}} = 1.00$, with negligible gate and channel coupling. It scores $K_{\text{catalytic}} = 6.722 \times 10^{-1}$ and Notch retention 13.31%, well below the 92% floor, and $\text{CKI}_{\text{TM}} = 0.0344$, two orders of magnitude below the potency threshold. Its catalytic leakage is 0.4436. In quadrant terms it is a clean IV. Its effect on the pathogenic ratio is $1.004\times$, that is, none: an inhibitor turns the enzyme down without changing what it makes. That single number is the phase III failure mode stated in the model’s own language.

Flurbiprofen stands for the first generation of NSAID-derived modulators: no meaningful active-site contact, weak gate and channel coupling. It is safe, $K_{\text{catalytic}} = 7.342 \times 10^{-5}$ and retention 99.98%, and silent, $\text{CKI}_{\text{TM}} = 0.2194$, giving a $1.021\times$ shift in the ratio. Quadrant III.

KMS-AD-309 is a designed point: full gate and channel coupling, $s_{\text{cat}} = 0.02$. It satisfies both constraints, $K_{\text{catalytic}} = 1.688 \times 10^{-3}$ against 0.025 and $\text{CKI}_{\text{TM}} = 4.2536$ against 3.20, with retention 99.49% and $S_{\text{Notch}} = 1583$. Its covariance term is 95.11% eigenvector rotation: it rearranges the channel’s modes rather than rescaling their spectrum, which is the structural signature Paper II associated with cryptic allostery. Its step weights are anisotropic but only mildly so, (1.3022, 1.2097, 1.2750), and the resulting discount is $0.903 k_B T$ against the isotropic ideal $0.751 k_B T$ at the apo weights. The ratio falls to 0.06978, a $2.61\times$ reduction.

4.3. A screen of 240 ligands and the conventional read-out. Sixty ligands were sampled for each quadrant by drawing the three occupancy strengths from ranges chosen to realise that quadrant; Table 6 reports the resulting spans. Exactly the sixty quadrant-II ligands satisfy both design constraints, and no others.

Now consider a conventional cell-based γ -secretase activity read-out, which sees the catalytic block and nothing else: it calls a ligand a hit when $K_{\text{catalytic}} > 0.025$. Of the 240 ligands it returns 120 hits, its recall on quadrant II is 0 out of 60, and every hit lies in quadrant I or IV, that is, carries a Notch liability.

Proposition 4.2 (What is definitional and what is not). *Let the quadrant partition be defined by the same threshold $\theta = 0.025$ that the assay uses. Then:*

- (i) (Definitional.) *The assay hit set is exactly $I \cup IV$, so its recall on quadrant II is 0 and its precision for the label “Notch-liable” is 1. No computation is involved.*
- (ii) (Empirical.) *That quadrant II is non-empty, and indeed well populated at realistic coupling strengths, is not definitional. It is a property of the model, and by Theorem 2.3 it is a consequence of the weakness of K_{CM} . Here 60 of 240 sampled ligands, a quarter of the library, meet both constraints.*
- (iii) (Empirical.) *None of the 120 assay hits satisfies both design constraints, although 60 of them have $CKI_{TM} > CKI^*$. Potency in the second-order index is therefore not what the assay is failing to see; safety is.*

Proof. (i) is the definition of the quadrants. (ii) and (iii) are read from Table 6. $\square$

We spell this out because the headline version of the observation, that the standard assay has zero recall on the useful class, is circular if stated without (ii) and (iii). The substantive claim is the conjunction: the target class exists, it is reachable by the Schur mechanism, it is a quarter of the library, and the read-out in routine use partitions the library along an axis orthogonal to the one that matters, enriching for exactly the compounds whose clinical history is worst.

Two features of Table 6 are worth reading directly. First, quadrants I and II are indistinguishable in the potency coordinate: their CKI_{TM} spans are $[4.071, 7.001]$ and $[4.308, 6.640]$, overlapping almost completely. Potency does not separate the safe from the unsafe here, and no amount of optimising it will. Second, the pathogenic ratio tracks the potency coordinate and ignores the safety coordinate exactly as (3.8) says it should: quadrants I and II share the range $R_{42/40} \in [0.024, 0.074]$, quadrants III and IV share $[0.143, 0.180]$. A screen that ranked compounds by predicted ratio alone would therefore rank the Notch-liable quadrant I ligands at the top, alongside the target class and slightly ahead of it. The safety coordinate is not a tie-breaker to be applied late; it is one of the two axes.

4.4. A design recipe. The results assemble into a procedure that does not depend on the particular constants used here.

- (1) *Partition the target.* Identify the catalytic block $\mathcal{C}$ whose geometry must be preserved, the machinery $\mathcal{M}$ whose covariance is to be rearranged, and measure the coupling block K_{CM} . Theorem 2.3 makes the size of that block the single most informative number about whether a selective allosteric ligand is possible at all: the achievable separation grows like $\|K_{CM}\|^{-2}$.
- (2) *Fix the safety threshold from the biology, not the chemistry.* Here the requirement “Notch retention above 92 %” fixes $K_{catalytic} < 0.025$ through (2.4). The threshold is a property of the off-target pathway and its tolerance, and it should be set before any ligand is scored.
- (3) *Score on both axes and never on their ratio alone.* By Proposition 4.1 a single ratio saturates and stops carrying safety information precisely in the regime one is designing for. Report $K_{catalytic}$, CKI_{TM} and S_{Notch} together.
- (4) *Require rotation, not stiffening.* By Section 2.5 the commuting part of the covariance term is a poor route to the threshold; demand D_{rot}/D_{cov} large, which is a spectral test that costs nothing once the index has been computed.
- (5) *Spread the cooperativity.* By Numerical observation 3.4 the geometric discount on the collective step is maximised when the step coordinates are coupled isotropically. A ligand that grips one helix is paying almost the sequential price.

- (6) *Check the two routes against each other.* The ratio law and the cone barrier are different computations. Their quotient is a free diagnostic, and a value far from unity indicates that the phenomenological constants have drifted away from the geometry.

Steps (1), (3), (4) and (6) are consequences of results proved here; steps (2) and (5) depend respectively on a biological input and on a numerical observation.

5. NUMERICAL VERIFICATION TABLES AND FIGURE

All numbers below are produced by `code/exp09_ad_gamma_secretase_gsm.py` (which prints **OVERALL ACCEPTANCE: PASS** on eight criteria, runtime about ten seconds) and by the companion `code/paper09_compute_tables.py` (fifteen checks, all passing). The console logs are `results/exp09_console_log.txt` and `results/paper09_tables_summary.txt`.

TABLE 1. The three-zone model of Definition 2.1. The stiffness is a weighted graph Laplacian plus a tether $t = 0.9$; its spectrum is $[0.9000, 24.8622]$ with condition number 27.62 and $\dim \ker K = 0$, as Proposition 2.2 requires. Inter-block couplings: gate–channel 1.0, channel packing 1.2, catalytic–channel 0.35, catalytic–gate 0.25.

block	dof	segments (dof)	path stiffness
catalytic $\mathcal{C}$	30	D257 loop (8), D385 loop (8), S3 register (14)	6.0
gate $\mathcal{G}$	70	TM6a (40), PAL (30)	4.0
channel $\mathcal{H}$	200	TM1–TM9 (24, 24, 26, 22, 22, 24, 22, 20, 16)	5.0

TABLE 2. Theorem 2.3: the catalytic couplings are scaled by s and the perturbation is held fixed at the purely allosteric ligand. $\|\Delta P\|_F$ is the change of the catalytic precision (2.6); leakage is $\|\Delta \Sigma_{CC}\|_F / \|\Sigma_{CC}\|_F$. Asymptotic exponents fitted on $s \leq 0.125$: 1.9906, 1.9709, 1.9755, 3.9633, against the predicted 2, 2, 2, 4. Over the whole scan CKI_{TM} varies by a factor 1.0050.

s	$K_{\text{cat,chan}}$	$\ \Delta P\ _F$	leakage	D_{mean}	D_{cov}	CKI_{TM}
0.03125	0.01094	1.58×10^{-5}	9.19×10^{-7}	2.11×10^{-10}	7.64×10^{-13}	4.2654
0.06250	0.02187	6.32×10^{-5}	3.65×10^{-6}	8.38×10^{-10}	1.21×10^{-11}	4.2650
0.12500	0.04375	2.51×10^{-4}	1.45×10^{-5}	3.30×10^{-9}	1.90×10^{-10}	4.2641
0.25000	0.08750	9.94×10^{-4}	5.53×10^{-5}	1.28×10^{-8}	2.90×10^{-9}	4.2625
0.50000	0.17500	3.89×10^{-3}	2.07×10^{-4}	4.85×10^{-8}	4.24×10^{-8}	4.2593
1.00000	0.35000	1.49×10^{-2}	7.30×10^{-4}	1.74×10^{-7}	5.71×10^{-7}	4.2536
2.00000	0.70000	5.48×10^{-2}	2.34×10^{-3}	5.72×10^{-7}	6.70×10^{-6}	4.2441

6. ASSESSMENT, LIMITS AND OPEN PROBLEMS

6.1. **What was established.** Table 7 summarises the status of each claim: whether it is proved, and where it is checked.

6.2. **Limits.** We state five, in the order in which they would bite.

(a) **The dynamics are a harmonic surrogate.** The model is a Gaussian elastic network on a *schematic* contact graph with the correct block topology: the catalytic aspartates on TM6 and TM7, the TM6a/PAL gate, nine transmembrane helices. It is not built on a deposited structure, it is not molecular dynamics, and no docking or free-energy calculation appears anywhere. Every quantity reported is a property of Definition 2.1.

(b) **The compounds enter through their class, not their chemistry.** Semagacestat, Flurbiprofen and KMS-AD-309 are represented by the triple $(s_{\text{cat}}, s_{\text{gate}}, s_{\text{chan}})$ and by nothing

TABLE 3. Theorem 3.2 and Numerical observation 3.4: the anisotropy family (3.5) at fixed $\sum_j w_j = 3.15 k_B T$. The relative cost rises and the discount falls monotonically away from the isotropic point. Distances in $k_B T$.

a	w_{TM3}	w_{TM6a}	w_{PAL}	d_L^{cone}	$d_L^{\text{cone}} / \sum_j w_j$	discount
0.0	1.0500	1.0500	1.0500	2.39868	0.761486	0.75132
0.1	1.1550	1.0500	0.9450	2.40137	0.762341	0.74863
0.2	1.2600	1.0500	0.8400	2.40950	0.764921	0.74050
0.3	1.3650	1.0500	0.7350	2.42320	0.769270	0.72680
0.4	1.4700	1.0500	0.6300	2.44273	0.775468	0.70727
0.5	1.5750	1.0500	0.5250	2.46846	0.783639	0.68154
0.6	1.6800	1.0500	0.4200	2.50099	0.793966	0.64901
0.7	1.7850	1.0500	0.3150	2.54119	0.806727	0.60881
0.8	1.8900	1.0500	0.2100	2.59045	0.822364	0.55955

TABLE 4. The three reference compounds: occupancy strengths, the two quadrant coordinates, the rotation part of the covariance term, the selectivity index (4.1) and the quadrant. Retention is $100 e^{-3K_{\text{catalytic}}}$ per cent; the design floor is 92 % and the potency threshold is 3.20.

compound	s_{cat}	s_{gate}	s_{chan}	$K_{\text{catalytic}}$	ret. (%)	CKI_{TM}	D_{rot}	S_{Notch}	quad.
Semagacestat	1.000	0.05	0.05	6.722×10^{-1}	13.31	0.0344	0.0286	0.05104	IV
Flurbiprofen	0.004	0.18	0.12	7.342×10^{-5}	99.98	0.2194	0.1919	204.4	III
KMS-AD-309	0.020	1.00	1.00	1.688×10^{-3}	99.49	4.2536	4.0433	1583	II

TABLE 5. The same compounds in the cone geometry. Step weights from (3.4), the cone distance of Definition 3.1, the synchronous fraction (3.6), the barrier discount (3.7) in kcal/mol with the corresponding rate factor for the $A\beta_{42} \rightarrow A\beta_{38}$ step, and the ratio law (3.8) with its fold reduction against $R_0 = 0.182$. Weights and distances in $k_B T$.

compound	w_{TM3}	w_{TM6a}	w_{PAL}	$\sum_j w_j$	d_L^{cone}	σ	$\Delta\Delta G^\ddagger$	$k\text{-fold}$	$R_{42/40}$	fold
Semagacestat	1.1026	1.0794	1.0860	3.2680	2.4886	0.0107	0.0051	1.008	0.18136	1.004
Flurbiprofen	1.1568	1.1255	1.1506	3.4330	2.6142	0.0663	0.0334	1.056	0.17826	1.021
KMS-AD-309	1.3022	1.2097	1.2750	3.7869	2.8841	0.7353	0.4090	1.942	0.06978	2.608

TABLE 6. The screen of 240 ligands, sixty sampled for each quadrant. ‘‘Assay’’ counts hits of the catalytic activity read-out, $K_{\text{catalytic}} > 0.025$; ‘‘both’’ counts ligands meeting both design constraints. Retention in per cent.

quadrant	n	$K_{\text{catalytic}} \text{ min}$	$K_{\text{catalytic}} \text{ max}$	$\text{CKI}_{\text{TM}} \text{ min}$	$\text{CKI}_{\text{TM}} \text{ max}$	ret. min	ret. max	assay	both
I	60	2.04×10^{-1}	6.70×10^{-1}	4.071	7.001	13.39	54.30	60	0
II	60	4.29×10^{-6}	1.24×10^{-2}	4.308	6.640	96.34	100.00	0	60
III	60	1.96×10^{-5}	1.14×10^{-2}	0.115	1.553	96.63	99.99	0	0
IV	60	2.09×10^{-1}	6.71×10^{-1}	0.098	1.710	13.35	53.50	60	0

else. The model reproduces the clinical separation of an orthosteric inhibitor, a weak modulator and a designed modulator; it does not predict that separation from structure, and it could not, because the map from a molecule to its triple is exactly what is missing.

(c) **KMS-AD-309 is a designed point, not a compound.** It is a location in the parameter space of Definition 2.1 chosen to satisfy the design brief. No synthesis, no assay, no measurement.

TABLE 7. Status of the claims. “Proved” means proved here for the model of Definition 2.1; “numerical” means verified computationally and not proved.

claim	status	evidence
$\lambda_{\min}(K) = t$ exactly	proved	Prop. 2.2; 0.9000 measured
catalytic precision is bilinear in $K_{\mathcal{CM}}$	proved	Thm 2.3(i), Eq. (2.6)
leakage and mean term are $O(s^2)$	proved	Thm 2.3(ii)–(iii); 1.97, 1.98
covariance term is $O(s^4)$	proved	Thm 2.3(iii); 3.96
CKI _{TM} is screening-independent	proved	Thm 2.3(iv); factor 1.0050
equal-weight closed form	proved	Thm 3.2(ii); error 4.44×10^{-16}
$\max_j w_j \leq d_L^{\text{cone}} < \sum_j w_j$	proved	Thm 3.2(iii); 243/243
anisotropy family is even in a	proved	Prop. 3.3; error 0
isotropy minimises d_L^{cone} at fixed sum	numerical	Obs. 3.4; 2001 points
well-posedness of the recursion	numerical	Eq. (3.2); margin $\geq 7.2 \times 10^{-5}$
S_{Notch} saturates at CKI _{TM} / ε_0	proved	Prop. 4.1; max ratio 0.9957
assay recall on quadrant II is zero	definitional	Prop. 4.2(i)
quadrant II is non-empty and reachable	numerical	Prop. 4.2(ii); 60/240
two routes agree to a factor	numerical	§3; quotient 1.343

(d) **Two constants are calibrated.** In (3.8), $R_0 = 0.182$ and $\eta = 0.45$ are phenomenological. The agreement between the ratio law and the cone barrier, quotient 1.343, constrains them but does not determine them, and a different calibration would move the fold reductions of Table 5 while leaving the quadrant assignments, which depend on neither constant, untouched.

(e) **The interesting number is the zero, not the hundred per cent.** The screen’s 100 % quadrant agreement says only that the sampled strength ranges realise their intended quadrants; it is a design check on the sampler, not a validation of any classifier against experiment. The informative statement is Proposition 4.2, and within it the conjunction of (ii) and (iii) rather than (i).

6.3. Open problems.

- (1) *Schur-convexity of the cone distance.* Prove or disprove that d_L^{cone} is a convex function of w on $(0, \infty)^k$. By symmetry this would upgrade Numerical observation 3.4 to a theorem valid for every direction, not merely along the family (3.5), and would establish isotropic optimality of the cooperativity discount in general.
- (2) *The degenerate regime of the recursion.* Definition 3.1 presupposes (3.2). Characterise the set of weights where it fails, and extend the recursion by replacing the summand with its positive part. The family (3.5) approaches the boundary as $a \rightarrow 1$, where the margin falls to 7.2×10^{-5} .
- (3) *Calibrating the ratio law.* Determine η and R_0 from cell-based A β profiling of a compound series with independently measured channel rearrangement, rather than fitting them. A single series spanning the threshold CKI* would suffice to test the exponential form itself.
- (4) *Real ensembles.* Rebuild Definition 2.1 on a deposited γ -secretase structure with molecular-dynamics ensembles, using the estimators of Papers VII–VIII, and test whether the Schur exponent of Theorem 2.3 survives contact with a real coupling block.
- (5) *The missing map.* Learn the map from a molecule to $(s_{\text{cat}}, s_{\text{gate}}, s_{\text{chan}})$. Without it the framework classifies ligands it is given but cannot propose them, and Limit (b) remains the binding constraint on the whole programme.

Data and code availability. Everything in this paper is produced by two scripts in `code/`:

- `exp09_ad_gamma_secretase_gsm.py`, the experiment;

- `exp09_lib.py`, its model library;
- `paper09_compute_tables.py`, the companion that writes the table bodies.

The three data files are in `data/` and the logs in `results/`. The package `kakutani_pharma` is inherited unchanged from Papers VII–VIII. Fourteen model unit tests and the twelve inherited package tests pass.

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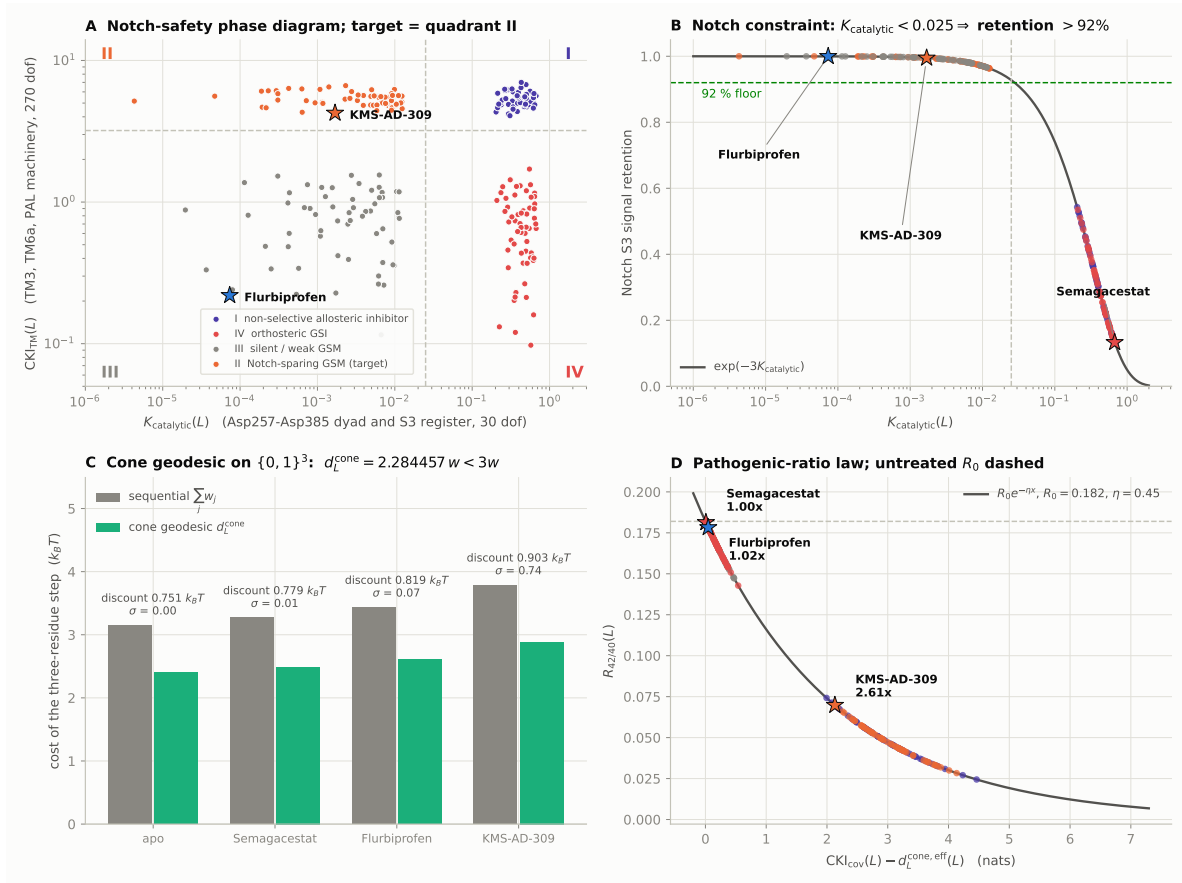


FIGURE 1. (A) The Notch-safety phase diagram: the 240 screened ligands and the three reference compounds in the plane of Section 1, with the thresholds $K_{catalytic} = 0.025$ and $CKI_{TM} = 3.20$ dashed. Quadrant II is the target class. (B) The Notch constraint (2.4): retention against $K_{catalytic}$, with the 92 % floor. Semagacestat falls to 13.31 %; both modulators sit at the plateau. (C) The cone geodesic of Definition 3.1 against the sequential cost, for the apo model and the three compounds, with the discount and the synchronous fraction σ . (D) The ratio law (3.8); the dashed line is the untreated $R_0 = 0.182$.